

Lab 6 Activity

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

1

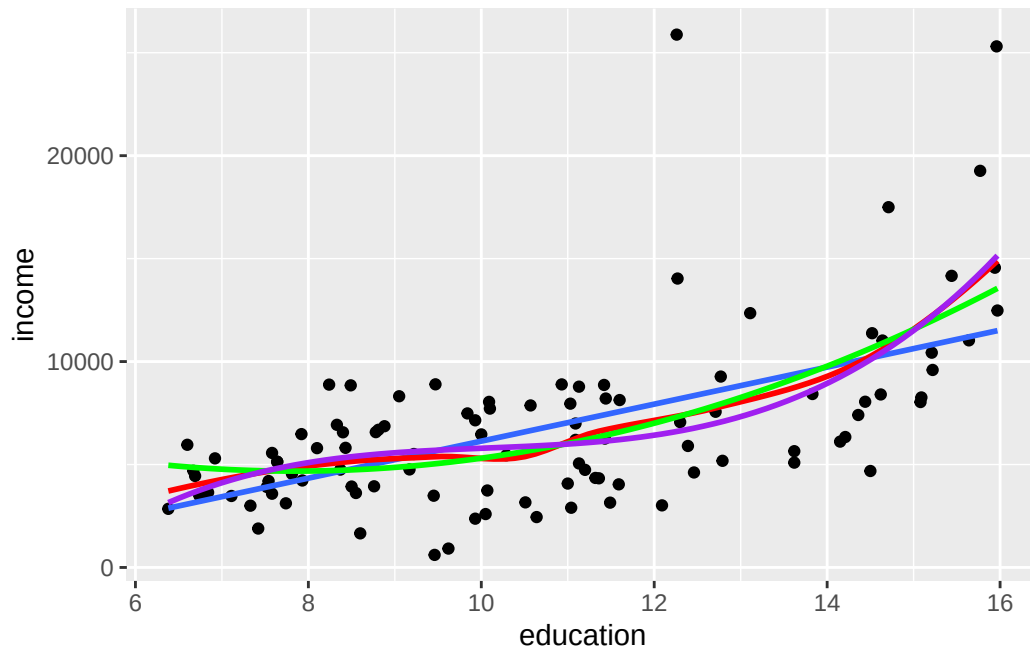
In this activity, we will work with the Prestige data in the carData package.

```
data("Prestige", package = "carData")
```

Look at the help page to better understand these data. Here, we want to predict the **income** of occupational incumbents from the average **education** of occupational incumbents. Use **ggplot** to create a scatterplot with one or more best-fit curves. What type of relationship do you think best characterizes the relationship between these variables?

```
data("Prestige", package = "carData")

ggplot(Prestige, aes(education, income)) +
  geom_point() + geom_smooth(method = lm, formula = 'y ~ x', se = FALSE) +
  geom_smooth(method = loess, formula = 'y ~ x', col = "red", se = FALSE) +
  geom_smooth(method = lm, formula = y ~ poly(x, 2), col = "green", se = FALSE) +
  geom_smooth(method = lm, formula = y ~ poly(x, 3), col = "purple", se = FALSE)
```



```
# quadratic might be a better descriptor than linear
```

2

Fit a simple linear regression model predicting income from education and evaluate assumptions. What do you conclude?

```
mod <- lm(income ~ education, Prestige)
summary(mod)
```

Call:

```
lm(formula = income ~ education, data = Prestige)
```

Residuals:

Min	1Q	Median	3Q	Max
-5493.2	-2433.8	-41.9	1491.5	17713.1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2853.6	1407.0	-2.028	0.0452 *

```
education      898.8      127.0    7.075 2.08e-10 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3483 on 100 degrees of freedom

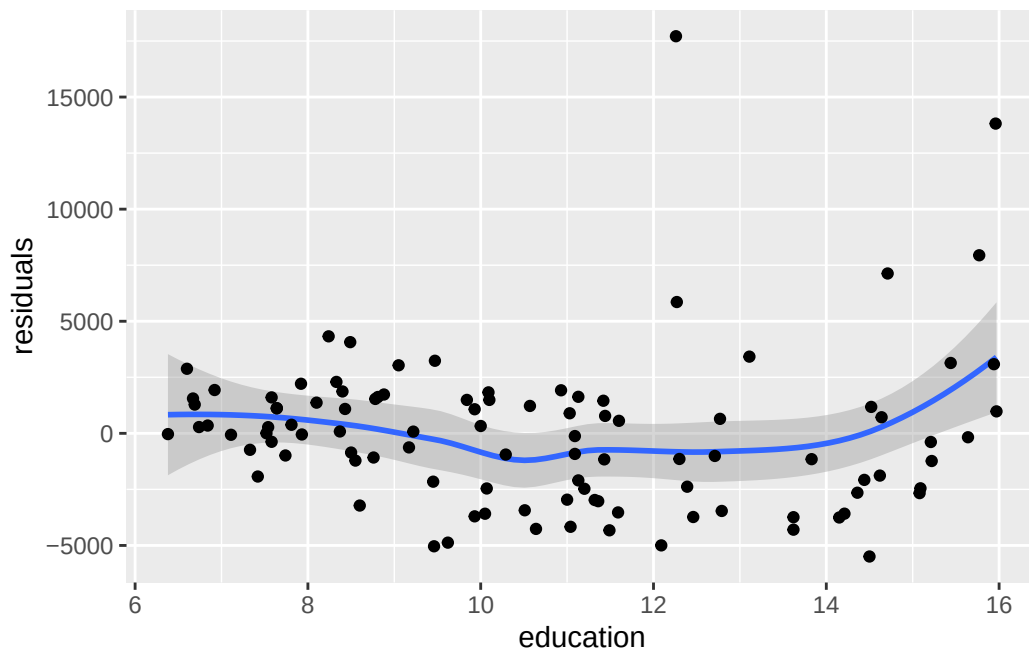
Multiple R-squared: 0.3336, Adjusted R-squared: 0.3269

F-statistic: 50.06 on 1 and 100 DF, p-value: 2.079e-10

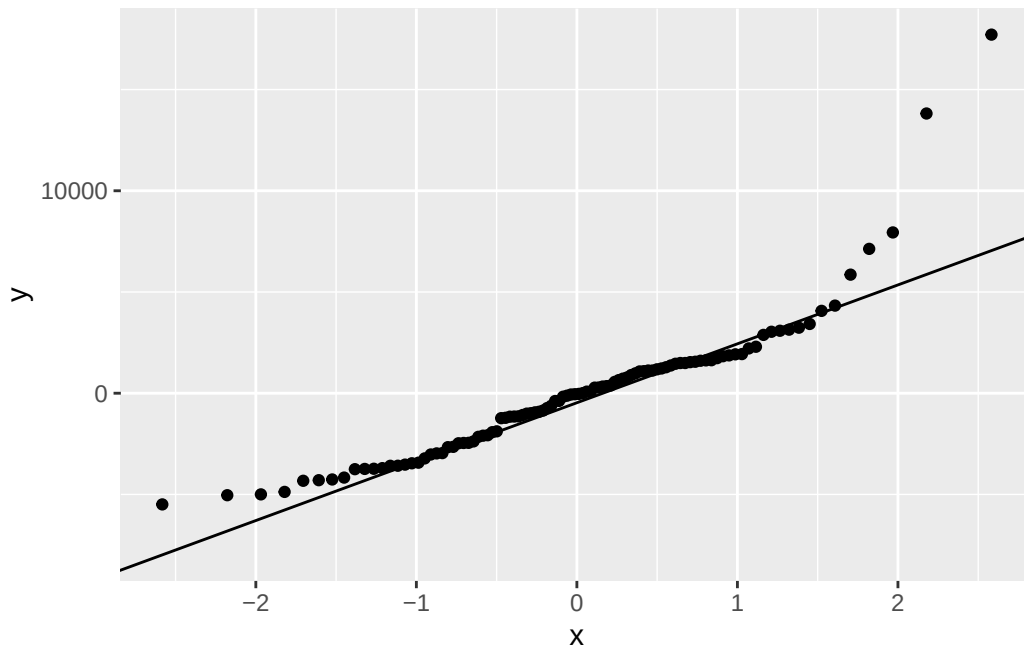
```
Prestige <- Prestige %>% mutate(residuals = residuals(mod))

ggplot(Prestige, aes(education, residuals)) +
  geom_smooth() + geom_point() # probably some heteroscedasticity
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



```
ggplot(Prestige, aes(sample = residuals)) +
  stat_qq() + stat_qq_line() # very non-normal
```



```
# there is a statistically significant relationship between education and
# income (R^2 = .33), but also some major violations of assumptions
```

3

Looking ahead: the `lm` function takes formula input like we used with `geom_smooth()`. Fit a regression equation for the best-fit quadratic/cubic curve predicting `income` from `education`. You can find the model residuals in the same way as we did for the linear model. Do the quadratic or cubic models do a better job meeting our assumptions about residuals? If not, can you think of what might be a better solution?

```
mod2 <- lm(income ~ poly(education, 2), Prestige) # quadratic
mod3 <- lm(income ~ poly(education, 3), Prestige) # cubic

summary(mod2)
```

Call:

```
lm(formula = income ~ poly(education, 2), data = Prestige)
```

Residuals:

Min	1Q	Median	3Q	Max
-5951.4	-2091.1	-358.2	1762.4	18574.2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6797.9	333.5	20.381	< 2e-16 ***
poly(education, 2)1	24645.9	3368.5	7.316	6.77e-11 ***
poly(education, 2)2	9488.3	3368.5	2.817	0.00586 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3369 on 99 degrees of freedom

Multiple R-squared: 0.383, Adjusted R-squared: 0.3706

F-statistic: 30.73 on 2 and 99 DF, p-value: 4.146e-11

```
summary(mod3)
```

Call:

```
lm(formula = income ~ poly(education, 3), data = Prestige)
```

Residuals:

Min	1Q	Median	3Q	Max
-5403.6	-2047.2	-345.9	1319.6	19275.9

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6797.9	328.5	20.696	< 2e-16 ***
poly(education, 3)1	24645.9	3317.4	7.429	4.1e-11 ***
poly(education, 3)2	9488.3	3317.4	2.860	0.00517 **
poly(education, 3)3	6698.0	3317.4	2.019	0.04621 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3317 on 98 degrees of freedom

Multiple R-squared: 0.4077, Adjusted R-squared: 0.3895

F-statistic: 22.48 on 3 and 98 DF, p-value: 3.693e-11

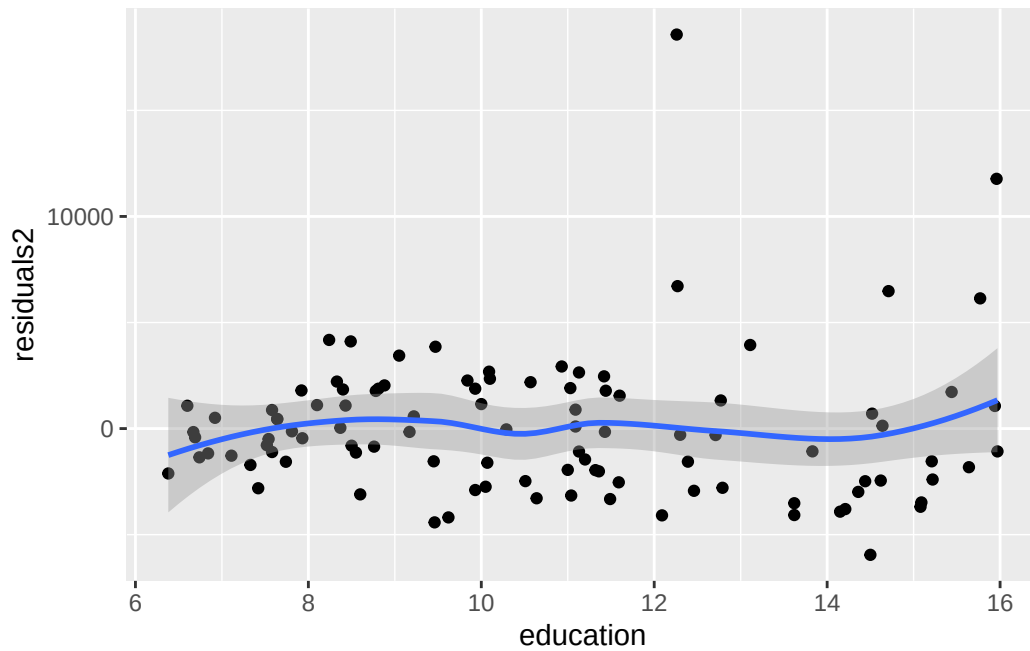
```
# both models have statistically significant R^2
# and both have higher R^2 than the linear model
```

```
Prestige <- Prestige %>%
```

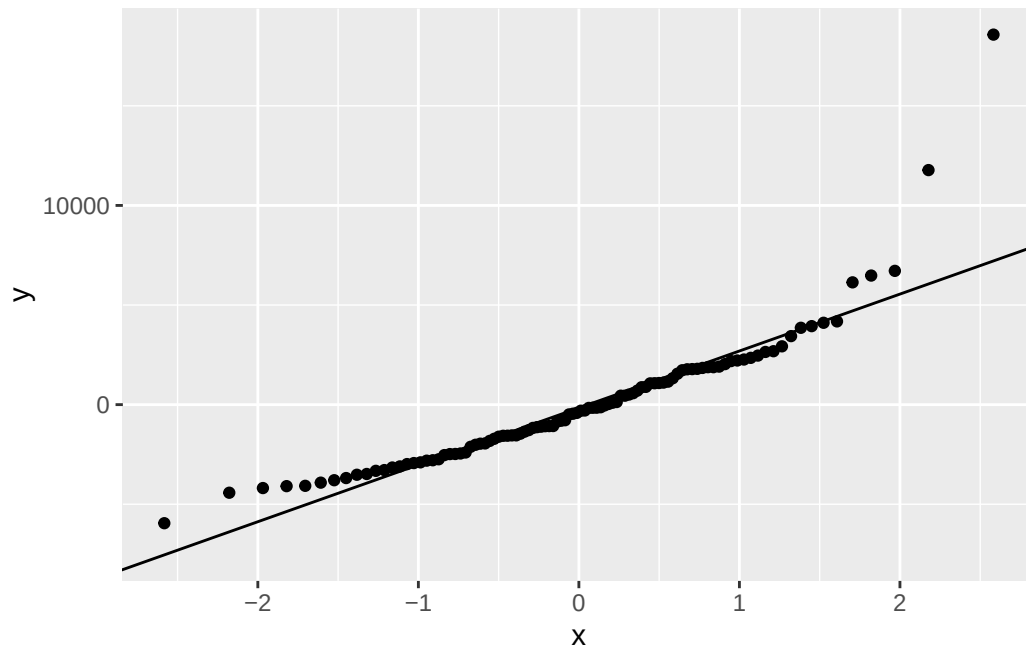
```
mutate(residuals2 = residuals(mod2),
       residuals3 = residuals(mod3))

ggplot(Prestige, aes(education, residuals2)) +
  geom_point() + geom_smooth(formula = y ~ x) # still heteroscedastic
```

`geom\_smooth()` using method = 'loess'

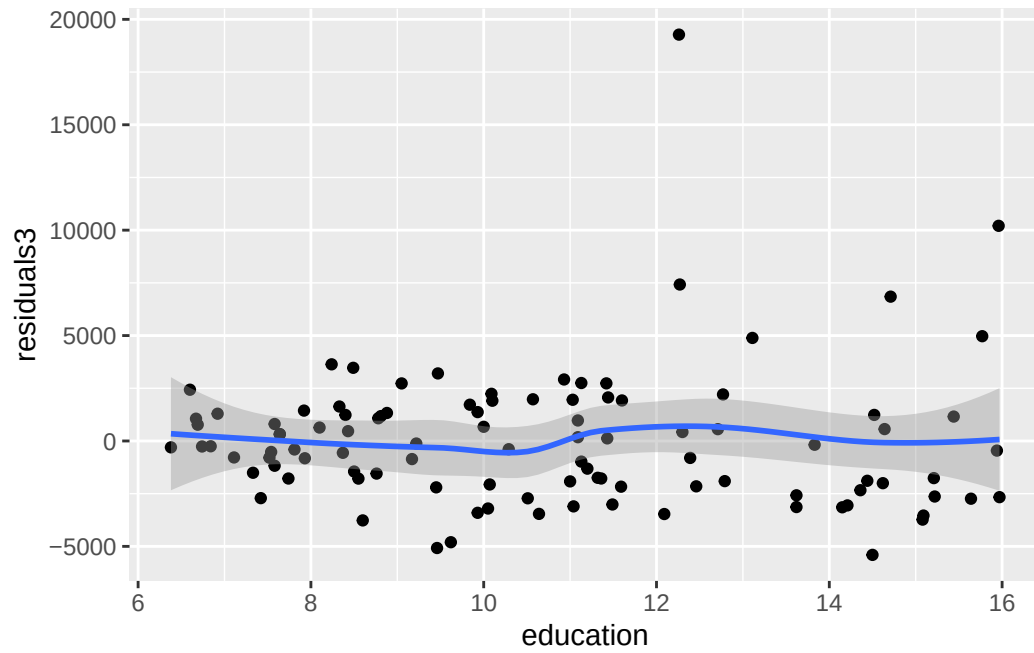


```
ggplot(Prestige, aes(sample = residuals2)) +
  stat_qq() + stat_qq_line() # still highly non-normal
```

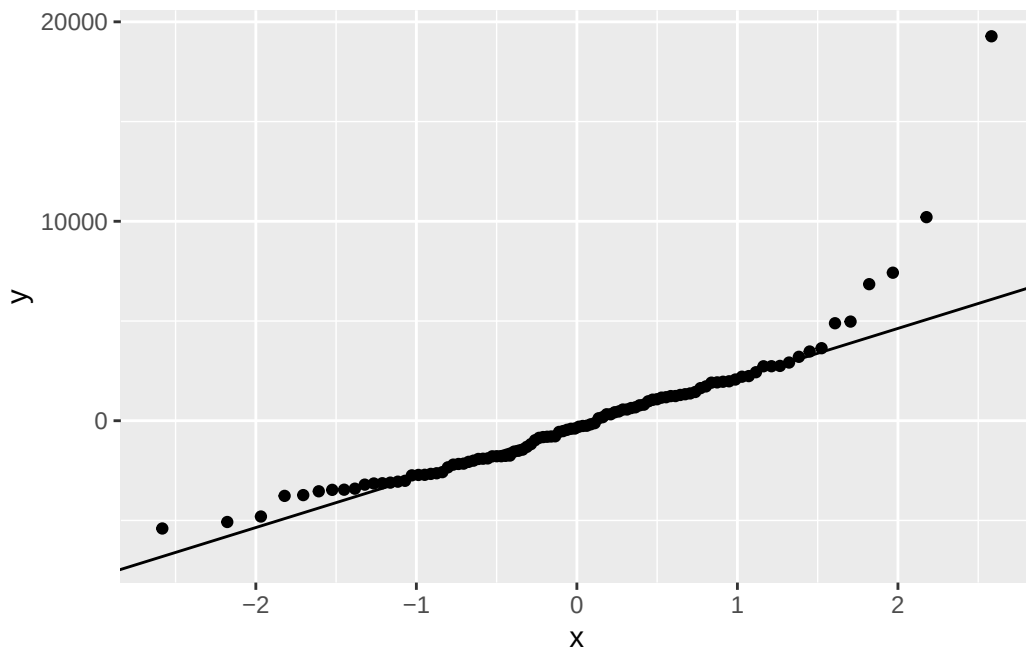


```
ggplot(Prestige, aes(education, residuals3)) +  
  geom_point() + geom_smooth(formula = y ~ x) # better, but there are still outliers
```

`geom\_smooth()` using method = 'loess'



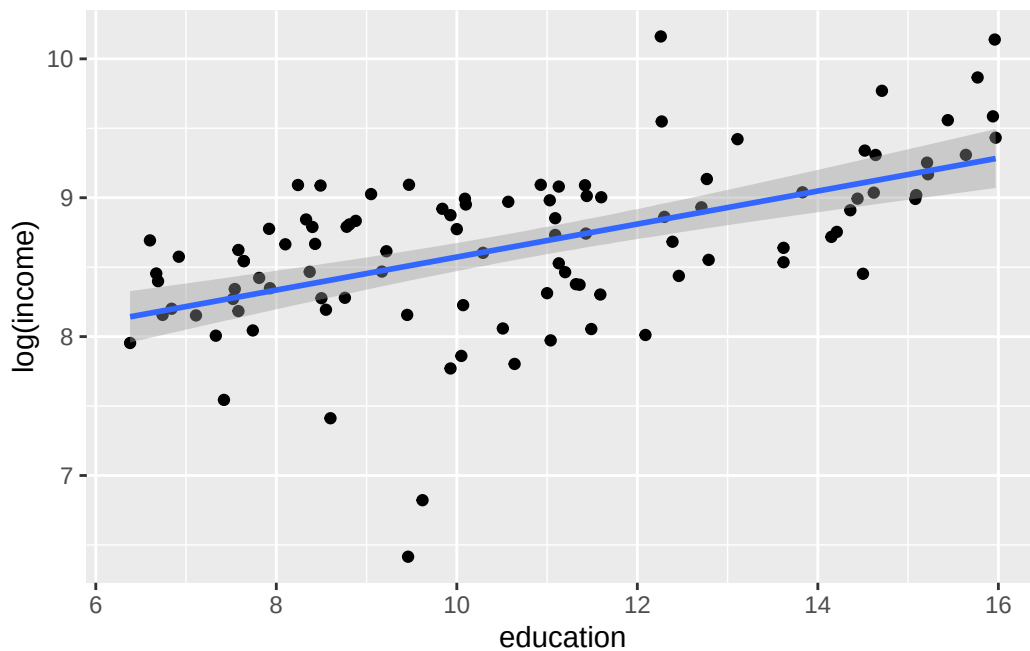
```
ggplot(Prestige, aes(sample = residuals3)) +
  stat_qq() + stat_qq_line() # still not great
```




```
# although the quadratic and cubic models do appear to perform better,  
# they still don't meet assumptions well  
# a better solution *might* be to log-transform the income variable:
```

```
ggplot(Prestige, aes(education, log(income))) +  
  geom_point() + geom_smooth(method = lm)
```

`geom\_smooth()` using formula = 'y ~ x'



```
# although there still are a couple outlying points, this does seem to  
# better meet the regression assumptions  
plot(lm(log(income) ~ education, Prestige))
```

